Full-Precision Benchmarking of Open Medical and General Language Models for a Tumour Board Assistant in India

Technical report

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Abstract

Vaidyx-TB is an assistant for hospital Tumour Boards in India. It checks, routes and summarises the records of a Case File and keeps every fact it shows tied to a verbatim quote from a source document; it does not interpret images, stage disease or advise treatment. Choosing the open-weight models behind it needs scores that are comparable with the published model cards and that reflect the exact weights the system runs. We benchmarked 13 open models (11 medical fine-tunes and 2 general models, 4B to 32B parameters, from developers in the USA, China and the UAE) on 11 public benchmarks, 5 text (9,995 questions) and 6 image (101,227 questions), with the published bf16 weights on one NVIDIA H200, greedy decoding, full test splits and no quantisation, under two protocols (step-by-step and direct answer): about one million graded answers. The harness reproduces published scores: MedGemma-27B-it lands within 2.5 points of its model card on all five text sets, and Lingshu-32B scores 82.4 % against a published 83.4 % on OmniMedVQA. Gemma-4-31B-it, a general model, is the strongest text model at 76.6 % overall and 55.1 % on MedXpertQA, where every medical model stays below 41 %. HuatuoGPT-3-27B is the best medical model (71.2 %) and leads OmniMedVQA (82.9 %). Qwen3.5-4B matches MedGemma-27B-it on text at a seventh of the size, and Lingshu-32B leads radiology and pathology VQA. Full precision removes most of the gap seen in an earlier 4-bit run (older fine-tunes gain 11 to 30 points). Bootstrap intervals and paired tests show which differences hold: ranks 3 to 6 on text are statistically tied, and the top two lead firmly, also on the 674 questions that mention cancer. We then report the grounded evaluations that decide what Vaidyx-TB ships: a reader and extractor bake-off on synthetic Case Files; 476 questions to the Case File with no fabricated value shown, where an ablation shows that the reasoner itself rarely writes an unsupported statement (the grounding layer removed five, two of them with a wrong detail); 10,883 values read from 2,102 real TCGA pathology reports, of which 1.4 % could not be tied to a quote and are flagged, with agreement with the clinical record of 90 to 98 % for histology, pT, pN, ER and PR and 84 % for HER2, the disagreements almost all stated in the reports themselves; report-status detection on 9,523 real reports; a summary omission check; and an independent OCR number check (98.6 % number recall; 99.4 % of planted digit substitutions caught with no false alarm). Every answer is saved per question, so every number here can be re-scored.

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1 Introduction

A Tumour Board (multidisciplinary tumour board, MDT) decides the treatment plan of a cancer patient from the records of several departments. In India most of that preparation is manual. A 2026 survey of Indian cancer centres reports that 88.3 % keep Board records on paper or in mixed form, that 53.3 % have no radiologist taking part in the Board, and that 48.2 % have no system to monitor whether the recommended treatment was given [Sarveswaran et al., 2026].[2] Vaidyx-TB, built at TANUH, addresses the preparation: it reads the documents of a Case File (pathology, radiology, laboratory and discharge reports, scanned or typed), checks the hospital's pre-authorisation requirements, extracts the values the Board needs, routes requests to Pathology and Radiology, drafts the case summary and the specialty briefs, and records the decision and the follow-up. Its scope is fixed by design: it checks, routes, summarises and flags. It never interprets an image, computes a stage or advises a treatment, and every value or sentence it shows as fact carries a verbatim quote from the source document; anything that cannot be verified is dropped or marked for review.

The system runs open-weight models on a hospital's own server. Two questions decide which models: how much medical knowledge a model has, measured on public examinations, and how reliably it stays grounded in the documents it is given, which no public benchmark measures. This report answers the first question with a benchmark that we could trust to be like for like with the published model cards, and the second with Vaidyx's own evaluations.

The benchmark differs from the usual model-card table in four ways. (1) Every model runs with its exact published bf16 weights; no quantisation or other lossy step sits anywhere in the pipeline, because the deployed system also runs bf16. (2) Every dataset is run in full; nothing is sub-sampled. (3) Every model runs under the same two protocols, a step-by-step protocol that matches how model cards report their scores and a direct-answer protocol that measures knowledge without reasoning and the discipline to follow a strict output format. (4) Every answer is saved per question, so every score can be re-checked or re-scored from the saved replies without regenerating anything. An earlier internal run of seven of these models used 4-bit weights (Q4_K_M) on a small workstation; this report shows how much of that run's gap was the compression.

Our contributions are:

- a like-for-like benchmark of 13 open medical and general models on 11 public text and image benchmarks (111,222 questions, about one million graded answers), with the protocol, the scoring rules and the comparison with every published number we could find (Sections 2 to 7);
- evidence that the harness reproduces published scores where the protocol matches, and a precise account of where it does not and why (Tables 5 and 13);
- a measurement of what full precision changes against a 4-bit run of the same weights (Table 6);
- the grounded evaluations that select Vaidyx-TB's reader, extractor and reasoner, measured on synthetic Case Files and on real pathology reports (Section 8).

2 Models

Table 1 lists the 13 models. Each row carries two labels that appear in every result table of this report and in the Vaidyx-TB user interface: the developer's country, and whether the model is a medical fine-tune or a general model that is not medically validated. Seven models read images; the other six are text only and appear in the text tables only. Seven were part of the earlier 4-bit report. All weights are the published bf16 checkpoints from Hugging Face. Three models (HuatuoGPT-3, Baichuan-M2 and Qwen3.5) have a thinking mode; it was switched off, so that every model sees the same visible step-by-step prompt and the same token budget. Two models of the earlier report, a medical mixture-of-experts model and Asclepius-8B, were not repeated because no bf16 checkpoint was in this queue.

[2] Two of the three figures are derived from the survey rather than stated in it: 88.3 % is 63.5 % physical plus 24.8 % mixed records, and 53.3 % is the complement of the 46.7 % radiologist participation it reports.

Table 1: The 13 models. Type: Medical = fine-tuned for medicine; Non-medical = a general model, not medically validated. Earlier report = part of the 4-bit run of September 2026.

Model	Params	Developer	Country	Type	Reads images	Earlier report
Gemma-4-31B-it	31B	Google DeepMind	USA	Non-medical	Yes	New
HuatuoGPT-3-27B	27B	FreedomIntelligence, CUHK-Shenzhen	China	Medical	Yes	New
Qwen3.5-4B	4B	Alibaba Qwen team	China	Non-medical	Yes	New
MedGemma-27B-text-it	27B	Google	USA	Medical	No	New
MedGemma-27B-it	27B	Google	USA	Medical	Yes	Yes
Baichuan-M2-32B	32B	Baichuan AI	China	Medical	No	New
Lingshu-32B	32B	Alibaba DAMO Academy	China	Medical	Yes	New
MedGemma-4B-it	4B	Google	USA	Medical	Yes	Yes
Med42-8B (v2)	8B	M42 Health	UAE	Medical	No	Yes
JSL-MedLlama-3-8B-v2.0	8B	John Snow Labs	USA	Medical	No	Yes
MedGemma-1.5-4B-it	4B	Google	USA	Medical	Yes	Yes
OpenBioLLM-8B	8B	Saama AI Labs	USA	Medical	No	Yes
MedLlama2-7B	7B	Community fine-tune of Llama 2	USA	Medical	No	Yes

3 Benchmarks

Table 2 gives a data card for each of the 11 benchmarks: five text sets (9,995 questions) and six image sets (101,227 questions). All are run on their full test split; nothing is sub-sampled. MedQA, MedMCQA, PubMedQA and MMLU-Med are the four text sets used on the MedGemma, Med42 and OpenBioLLM model cards, which makes the comparison in Table 5 possible. MedMCQA, drawn from the Indian postgraduate entrance examinations, is the most India-relevant text set and the largest, so it carries the most weight in the overall text score, which is the accuracy over all 9,995 questions together. MedXpertQA is filtered so that strong models fail it; its ten options put chance at 10 %. The six image sets mix closed questions (yes/no, 2 to 5 options) and open short answers; OmniMedVQA alone has 88,996 questions over 42 public datasets and eight modalities.

4 Protocol

Hardware and engine. One NVIDIA H200 NVL (141 GB), vLLM 0.30 [Kwon et al., 2023], PyTorch 2.13, CUDA 13. One model on the GPU at a time, 90 % of memory, prefix caching on. The GPU ran power-capped at 300 W after earlier hardware faults; the cap slows runs but changes no answer, since decoding is greedy.

Weights and decoding. Exact published bf16 weights; no quantisation or other lossy compression anywhere. Greedy decoding (temperature 0), seed 0. Each model uses its own chat template; thinking mode off where a switch exists.

Two protocols. Under the *step-by-step* protocol (chain of thought, CoT) the instruction is: "Answer the following multiple-choice question. Think through the question step by step, then finish with a final line of the form 'Answer: X' where X is the letter of the correct option." The model may generate up to 2,048 new tokens, and the answer is read from the final `Answer:` line. This is how model cards report their results, so these are the scores to compare with published numbers. Under the *direct* protocol the instruction is: "Answer the following multiple-choice question. Respond with only the letter of the correct option.", the prompt ends with `Answer:`, and the model may generate at most 32 new tokens (yes/no, or a short phrase for open image questions). It measures knowledge without reasoning and, at the same time, whether the model follows a strict output format. The question and its options follow the instruction in the same user turn, through each model's own chat template.

Scores. The *strict* score counts an answer only if a valid option can be read from the reply; a model that starts explaining and is cut off at 32 tokens scores zero for that question even if it was heading to the right answer. The *first-token* score is a format-free check on the direct protocol: the option letter with the highest probability among the top-20 candidates for the first generated token. It shows what the model knew when the strict score is dragged down by formatting. *Parsed %* is the share of replies from which an answer could be read; below about 95 % a score is partly a formatting result. For VQA-RAD, SLAKE and PathVQA the score is split into closed (yes/no) and open short-answer questions; open answers are scored by exact match after normalising case and punctuation, which is strict and penalises every model equally. *Errors* are

Table 2: Data cards of the 11 benchmarks. Text sets above the rule (9,995 questions), image sets below (101,227 questions). Chance = the score of random guessing.

Dataset	What it tests	Questions	Format	Chance
MedQA	US medical licensing exam (USMLE) style clinical vignettes: diagnosis, next step, mechanism	1,273 (test)	4 options	25 %
MedMCQA	Indian postgraduate medical entrance exams (AIIMS, NEET-PG) across 21 subjects; the most India-relevant text test	4,183 (validation; test answers are hidden)	4 options	25 %
PubMedQA	Reading a research abstract and answering a yes / no / maybe question about it (evidence reading, not recall)	1,000 (expert-labelled)	yes / no / maybe	33 %
MMLU-Med	Six medical subjects of MMLU: anatomy, clinical knowledge, college biology, college medicine, medical genetics, professional medicine	1,089 (test)	4 options	25 %
MedXpertQA	Expert-level, deliberately hard questions from 17 specialty board exams, filtered so that strong models fail them; text subset	2,450 (test)	10 options	10 %
VQA-RAD	Radiology images (CT, MRI, X-ray) with clinician-written questions	451 (test)	yes/no and short answer	varies
SLAKE	Radiology images with organ, position and abnormality questions (English subset)	1,061 (test)	yes/no and short answer	varies
PathVQA	Pathology images and textbook figures; open answers are long medical phrases, so exact match is strict	6,719 (test)	yes/no and short answer	varies
PMC-VQA	Figures from PubMed Central papers (all modalities), cleaned test set	2,000 (test_clean)	4 options	25 %
MedXpertQA-MM	Expert board-exam questions that need one or more clinical images plus a vignette	2,000 (test)	5 options	20 %
OmniMedVQA	Open-access part of OmniMedVQA: 42 public datasets, 8 modalities (MRI, CT, ultrasound, X-ray, dermoscopy, microscopy, fundus, OCT); mostly modality, anatomy and disease recognition	88,996	2 to 4 options	about 27 %

questions the serving engine refused (an image too large for the model's context window, or more than four images); they count as wrong.

Published numbers. Every published score was opened directly from the official model card or paper (Section 12, pages opened on 9 October 2026). Where a card uses a different scoring rule, the rule is stated next to the number.

The earlier run. The previous internal report (September 2026) ran seven of these models as Q4_K_M 4-bit weights with Ollama on an NVIDIA GB10 workstation under a direct letter prompt, on the same five text splits. Its direct prompt is closest to this run's direct protocol.

Scale and safety of the run. 13 models under two protocols on text, 7 vision models on 6 image sets, plus the step-by-step protocol on the two multiple-choice image sets: about 1.0 million graded answers. OmniMedVQA alone took about 14 hours across the seven vision models. A thermal guard and GPU telemetry ran throughout. Every answer is saved per question (JSONL), so runs resume after an interruption and any score can be re-scored from the saved replies.

5 Text results

5.1 Step-by-step accuracy

Table 3 ranks the 13 models by their overall step-by-step accuracy. Gemma-4-31B-it, a general model, leads every column except PubMedQA: MedQA 93.2, MedMCQA 78.5, MMLU-Med 94.6, and 55.1 on

MedXpertQA, where every medical model stays below 41. HuatuoGPT-3-27B is the best medical model at 71.2 % overall and has the highest PubMedQA score (81.4). Qwen3.5-4B, a 4B general model, reaches 62.4 %, statistically tied with MedGemma-27B-it (61.7 %; Section 5.5) and within one point on MedQA (82.2 against 83.1). MedMCQA is the hardest of the four standard sets: the seven strongest models score 9 to 16 points lower on it than on MedQA. Two scores are held down by format rather than knowledge, which the last column shows: Med42-8B parses only 75 % of its PubMedQA replies and MedGemma-1.5-4B only 71 % of its MedQA replies.

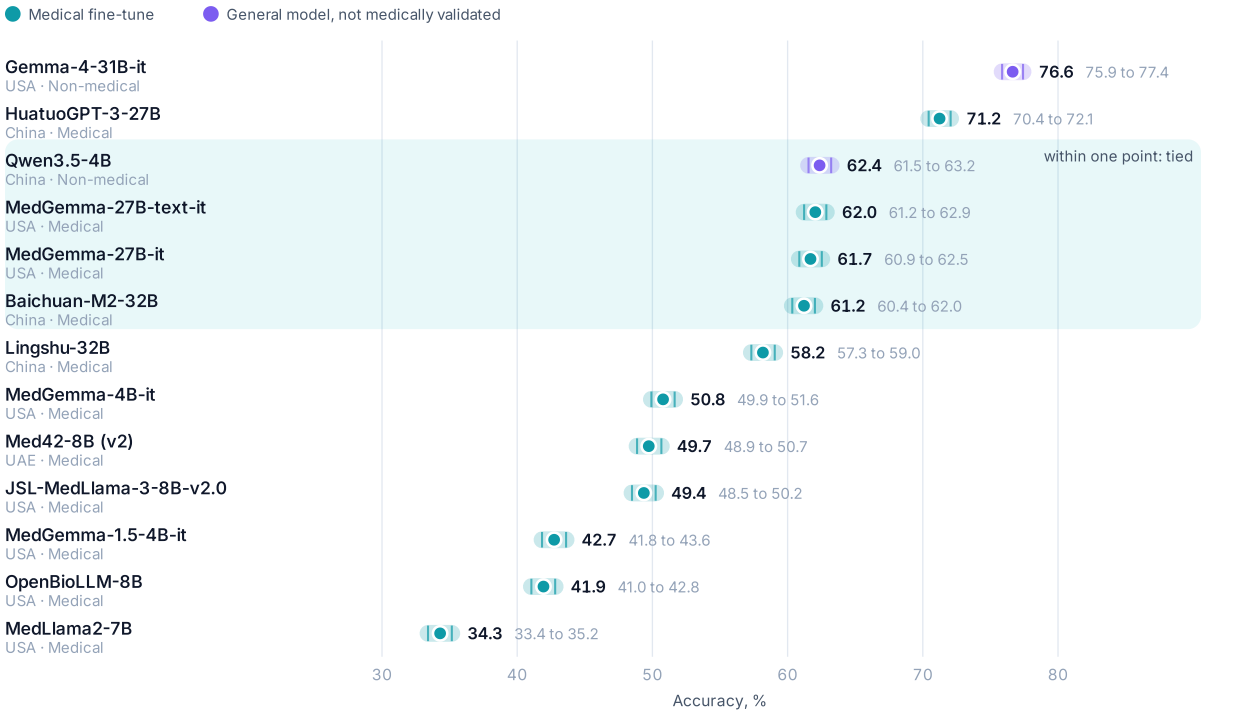


Figure 1: Overall step-by-step accuracy over all 9,995 text questions, with 95 % bootstrap confidence intervals (Section 5.5). The shaded models are statistically tied. Teal: medical fine-tunes; violet: general models, not medically validated.

Table 3: Step-by-step (CoT) accuracy in %, ranked by overall accuracy over all 9,995 text questions. Bold = best in column. Min parsed = the lowest parse rate across the five sets.

#	Model	Origin	Type	MedQA	MedMCQA	PubMedQA	MMLU-Med	MedXpertQA	Overall	Min parsed %
1	Gemma-4-31B-it	USA	Non-medical	**93.2**	**78.5**	80.7	**94.6**	**55.1**	76.6	100.0
2	HuatuoGPT-3-27B	China	Medical	91.0	75.3	**81.4**	92.5	40.4	71.2	97.7
3	Qwen3.5-4B	China	Non-medical	82.2	68.0	78.7	88.4	24.3	62.4	96.9
4	MedGemma-27B-text-it	USA	Medical	80.2	71.5	74.7	88.9	19.3	62.1	81.8
5	MedGemma-27B-it	USA	Medical	83.1	68.5	76.9	87.6	21.2	61.7	92.4
6	Baichuan-M2-32B	China	Medical	81.3	69.4	68.7	88.9	21.4	61.2	99.9
7	Lingshu-32B	China	Medical	76.7	62.7	80.3	84.0	20.2	58.2	99.5
8	MedGemma-4B-it	USA	Medical	66.1	56.1	75.8	71.3	14.4	50.8	99.0
9	Med42-8B (v2)	UAE	Medical	62.7	58.6	57.9	73.7	13.8	49.7	74.9
10	JSL-MedLlama-3-8B-v2.0	USA	Medical	58.8	55.5	76.3	72.9	12.5	49.4	98.2
11	MedGemma-1.5-4B-it	USA	Medical	46.9	50.5	60.9	64.7	10.2	42.7	70.6
12	OpenBioLLM-8B	USA	Medical	49.9	48.0	52.6	67.5	11.8	41.9	91.4
13	MedLlama2-7B	USA	Medical	38.2	34.9	69.9	48.3	10.5	34.3	95.8

5.2 Direct answers and the gain from reasoning

Table 4 gives the direct-answer scores, the first-token score, the parse rate and the gain from reasoning (step-by-step minus direct, in points). HuatuoGPT-3-27B is the best direct-answer model (66.0 %), which matters

for a system that asks short, structured questions. Reasoning helps the general models most: Gemma-4-31B-it gains 17.4 points overall and 44.1 points on MedXpertQA (11.1 to 55.1), and Qwen3.5-4B gains 9.3 points. Two Google models cannot follow the letter-only instruction: MedGemma-27B-text-it scores 8.0 on MedQA under the direct protocol with 25 % of replies parseable, against 80.2 with reasoning, and MedGemma-1.5-4B behaves the same way; their first-token scores (46.7 and 35.7 overall) show what they knew. Baichuan-M2's direct replies all began with a stray `</think>` tag that the parser rejected; the saved replies were re-scored with the tag removed, and nothing was regenerated (dagger in the table).

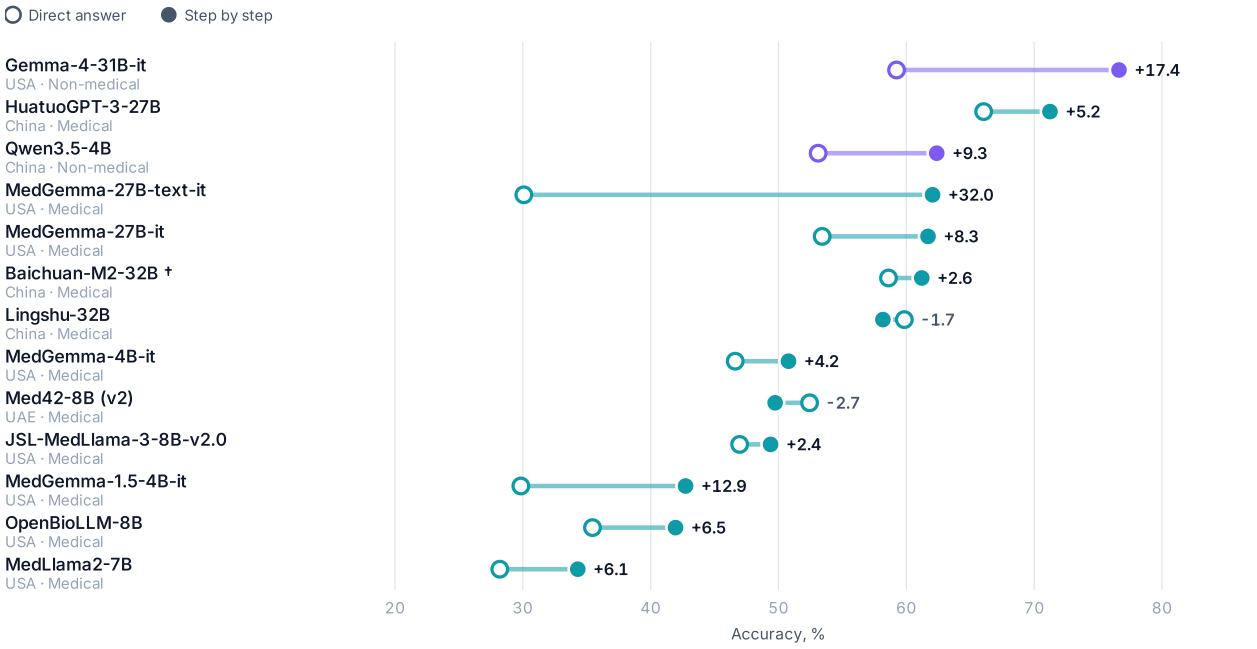


Figure 2: What reasoning adds: overall text accuracy with a one-letter answer (open dot) and step by step (filled dot); the number is the gain in points. †Re-scored from the saved replies.

Table 4: Direct-answer accuracy in % (at most 32 new tokens), ranked by overall. First-token = the format-free score. Parsed = the lowest parse rate across the five sets. CoT gain = overall step-by-step minus overall direct, in points. †Re-scored from the saved replies after removing a stray thinking tag.

Model	Origin	Type	MedQA	MedMCQA	PubMedQA	MMLU-Med	MedXpertQA	Overall	First-token overall	Parsed %	CoT gain
HuatuoGPT-3-27B	China	Medical	83.7	70.4	80.8	90.2	32.6	66.0	66.0	99.9	+5.2
Lingshu-32B	China	Medical	74.3	65.8	79.2	86.0	22.7	59.8	59.8	100.0	-1.7
Gemma-4-31B-it	USA	Non-medical	68.4	71.7	79.6	90.2	11.1	59.2	68.8	21.8	+17.4
Baichuan-M2-32B†	China	Medical	74.6	67.1	71.7	86.9	17.9	58.6	–	–	+2.6
MedGemma-27B-it	USA	Medical	66.3	60.7	77.2	81.0	12.3	53.4	55.5	70.4	+8.3
Qwen3.5-4B	China	Non-medical	67.6	58.0	76.5	79.0	16.2	53.1	53.1	100.0	+9.3
Med42-8B (v2)	UAE	Medical	62.1	60.1	77.2	72.9	15.1	52.4	52.5	99.9	-2.7
JSL-MedLlama-3-8B-v2.0	USA	Medical	45.0	56.5	77.2	67.9	10.0	47.0	47.8	74.5	+2.4
MedGemma-4B-it	USA	Medical	55.2	52.0	74.9	64.9	13.2	46.6	46.6	100.0	+4.2
OpenBioLLM-8B	USA	Medical	43.7	40.3	44.4	58.2	9.1	35.4	39.2	59.9	+6.5
MedGemma-27B-text-it	USA	Medical	8.0	38.2	72.3	49.0	2.0	30.1	46.7	14.2	+32.0
MedGemma-1.5-4B-it	USA	Medical	17.8	40.7	56.1	41.5	1.7	29.8	35.7	16.3	+12.9
MedLlama2-7B	USA	Medical	16.4	33.2	68.0	40.3	4.1	28.2	33.3	38.6	+6.1

5.3 Comparison with published scores

Table 5 sets our step-by-step scores against the published ones for the eight models that report any of these five sets. MedGemma-27B-it lands within 2.5 points of its card on all five sets (mean −1.1), MedGemma-4B-it slightly above its card (mean +1.2), and Lingshu-32B within 3.4 points on every set (mean −0.4); these three confirm that the harness is like for like with a generated-answer protocol. MedGemma-27B-text-it is

3.4 points below its card on average, mostly on MedQA and MedXpertQA. MedGemma-1.5-4B is 10 points below its card, mainly because 29 % of its MedQA replies never reach a final `Answer:` line within 2,048 tokens. Med42, JSL-MedLlama and OpenBioLLM are scored on their cards by answer likelihood with the lm-evaluation-harness [Gao et al., 2023], not by generated answers, which usually gives a few points more than a generated-answer test; OpenBioLLM is 12.3 points below its card on average and its PubMedQA gap is 21.5 points. No published scores on these sets exist for Gemma-4-31B, Qwen3.5-4B, HuatuoGPT-3-27B, Baichuan-M2-32B or MedLlama2-7B.

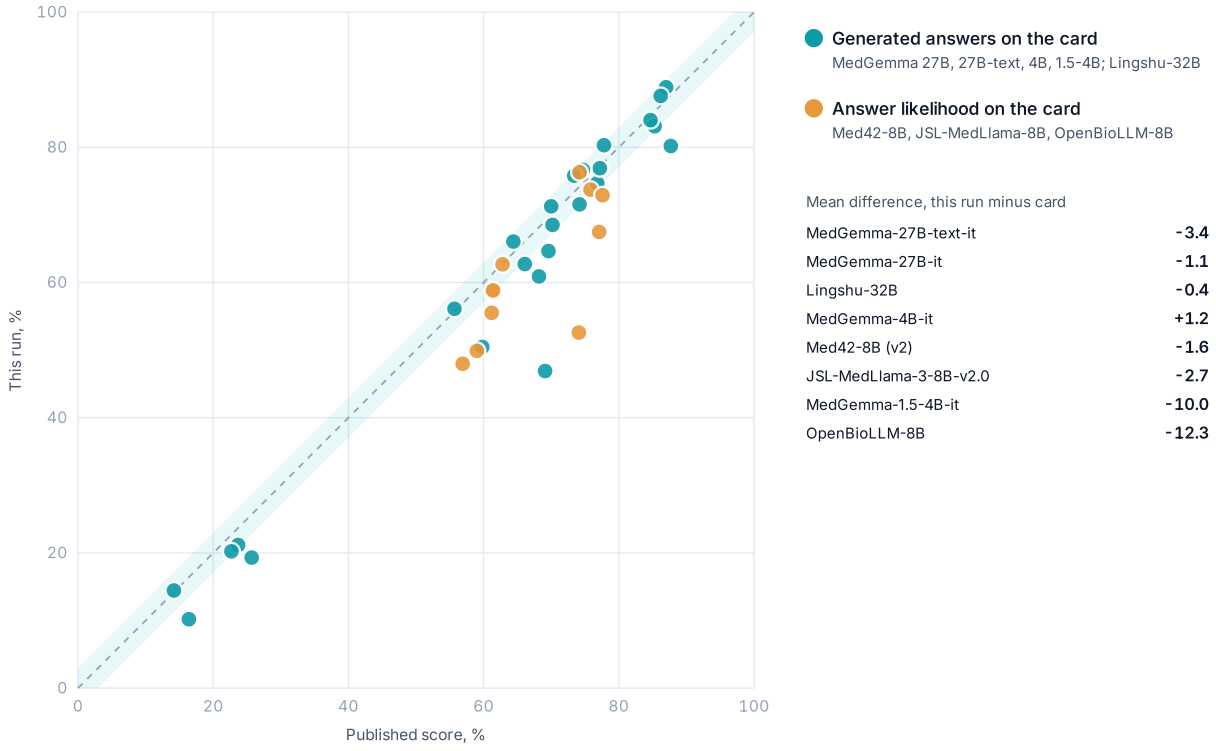


Figure 3: This run against the published model-card scores, one dot per model and text set; the dashed line is equality and the band is ±3 points. Orange: cards that score by answer likelihood rather than generated answers.

5.4 Full precision against the earlier 4-bit run

Table 6 compares the seven models of the September 2026 report, run then as Q4_K_M weights on a GB10 workstation, with this run on the same five splits. With the same weights at full precision and allowed to reason, MedGemma-27B-it rises from 69.8 to 83.1 on MedQA, and the older 7B to 8B fine-tunes gain between 10.6 and 30.1 points on average over the five sets (JSL-MedLlama +30.1, OpenBioLLM +20.6, MedGemma-1.5-4B +15.9, MedLlama2 +10.6). Under the direct protocol, which is closest to the earlier prompt, the gains are smaller and in two cases negative, so most of the earlier gap was the combination of 4-bit compression and a short-answer format, not the models themselves.

5.5 Uncertainty and ties

Every score above is an estimate from a finite set of questions. Table 7 gives 95 % percentile bootstrap intervals (4,000 resamples of questions, stratified by dataset for the overall score) and Table 8 the comparisons the text makes, as paired bootstrap intervals of the difference and McNemar's exact test on the questions where exactly one of the two models is right. An overall text score is uncertain by about ±0.8 points, a MedXpertQA score by about ±1.8, a MedQA score by ±1.3 to ±2.8. Ranks 3 to 6 (Qwen3.5-4B, MedGemma-27B-text-it, MedGemma-27B-it and Baichuan-M2) form one group within a point of each other that cannot

Table 5: This run (bf16, step-by-step) against the published scores, in %. Difference = this run minus published, in points; the mean is over the sets the source reports.

Model	Row	MedQA	MedMCQA	PubMedQA	MMLU-Med	MedXpertQA	Mean diff.
MedGemma-27B-text-it	This run (bf16, CoT)	80.2	71.5	74.7	88.9	19.3	
	Published (Google card)	87.7	74.2	76.8	87.0	25.7	
	Difference	-7.5	-2.7	-2.1	+1.9	-6.4	-3.4
MedGemma-27B-it	This run (bf16, CoT)	83.1	68.5	76.9	87.6	21.2	
	Published (Google card)	85.3	70.2	77.2	86.2	23.7	
	Difference	-2.2	-1.7	-0.3	+1.4	-2.5	-1.1
Lingshu-32B	This run (bf16, CoT)	76.7	62.7	80.3	84.0	20.2	
	Published (Lingshu report)	74.7	66.1	77.8	84.7	22.7	
	Difference	+2.0	-3.4	+2.5	-0.7	-2.5	-0.4
MedGemma-4B-it	This run (bf16, CoT)	66.1	56.1	75.8	71.3	14.4	
	Published (Google card)	64.4	55.7	73.4	70.0	14.2	
	Difference	+1.7	+0.4	+2.4	+1.3	+0.2	+1.2
Med42-8B (v2)	This run (bf16, CoT)	62.7	58.6	57.9	73.7	13.8	
	Published (M42 card[a])	62.8	61.3	–	75.8	–	
	Difference	-0.1	-2.7	–	-2.1	–	-1.6
JSL-MedLlama-3-8B-v2.0	This run (bf16, CoT)	58.8	55.5	76.3	72.9	12.5	
	Published (JSL card[b])	61.4	61.2	74.2	77.6	–	
	Difference	-2.6	-5.7	+2.1	-4.7	–	-2.7
MedGemma-1.5-4B-it	This run (bf16, CoT)	46.9	50.5	60.9	64.7	10.2	
	Published (Google card)	69.1	59.8	68.2	69.6	16.4	
	Difference	-22.2	-9.3	-7.3	-4.9	-6.2	-10.0
OpenBioLLM-8B	This run (bf16, CoT)	49.9	48.0	52.6	67.5	11.8	
	Published (Saama card[b])	59.0	56.9	74.1	77.1	–	
	Difference	-9.1	-8.9	-21.5	-9.6	–	-12.3

The Med42, JSL and Saama cards score by answer likelihood (lm-evaluation-harness), not by generated answers. [a]MMLU covers the clinical topics. [b]MMLU is the mean of the six medical subjects.

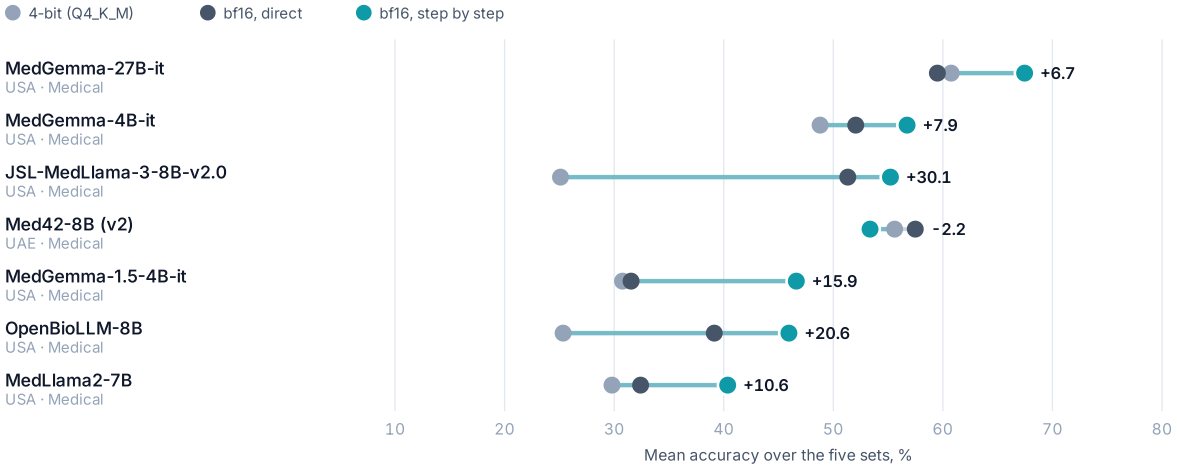


Figure 4: The same weights at 4 bits (September 2026) and at full precision, direct and step by step; mean over the five text sets. The number is the step-by-step gain over the 4-bit run.

Table 6: The earlier 4-bit run against this run, in %. Mean gain = the average of the five scores minus the Q4 average, in points.

Model	Run	MedQA	MedMCQA	PubMedQA	MMLU-Med	MedXpertQA	Mean gain
MedGemma-27B-it	Q4_K_M, GB10	69.8	62.1	76.1	79.2	16.6	
	bf16, H200, direct	66.3	60.7	77.2	81.0	12.3	-1.3
	bf16, H200, CoT	83.1	68.5	76.9	87.6	21.2	+6.7
MedGemma-4B-it	Q4_K_M, GB10	51.8	47.5	71.4	60.6	12.7	
	bf16, H200, direct	55.2	52.0	74.9	64.9	13.2	+3.3
	bf16, H200, CoT	66.1	56.1	75.8	71.3	14.4	+7.9
Med42-8B (v2)	Q4_K_M, GB10	60.4	57.9	75.6	71.7	12.4	
	bf16, H200, direct	62.1	60.1	77.2	72.9	15.1	+1.9
	bf16, H200, CoT	62.7	58.6	57.9	73.7	13.8	-2.2
JSL-MedLlama-3-8B-v2.0	Q4_K_M, GB10	27.6	31.0	36.0	21.6	9.3	
	bf16, H200, direct	45.0	56.5	77.2	67.9	10.0	+26.2
	bf16, H200, CoT	58.8	55.5	76.3	72.9	12.5	+30.1
MedGemma-1.5-4B-it	Q4_K_M, GB10	29.8	33.4	58.0	23.5	9.1	
	bf16, H200, direct	17.8	40.7	56.1	41.5	1.7	+0.8
	bf16, H200, CoT	46.9	50.5	60.9	64.7	10.2	+15.9
OpenBioLLM-8B	Q4_K_M, GB10	24.5	25.4	44.7	23.0	9.1	
	bf16, H200, direct	43.7	40.3	44.4	58.2	9.1	+13.8
	bf16, H200, CoT	49.9	48.0	52.6	67.5	11.8	+20.6
MedLlama2-7B	Q4_K_M, GB10	27.0	27.4	59.8	25.6	9.2	
	bf16, H200, direct	16.4	33.2	68.0	40.3	4.1	+2.6
	bf16, H200, CoT	38.2	34.9	69.9	48.3	10.5	+10.6

be separated: Qwen3.5-4B is +0.7 points above MedGemma-27B-it with an interval of −0.2 to +1.5 ($p = 0.13$), which is what "matches" means here. Med42-8B and JSL-MedLlama, and MedGemma-1.5-4B and OpenBioLLM-8B, are also tied. The leads of Gemma-4-31B-it (+5.4 over HuatuoGPT-3) and HuatuoGPT-3 (+8.9 over Qwen3.5-4B) are firm, and so are the gains and losses from reasoning. On images, HuatuoGPT-3's OmniMedVQA lead over Lingshu-32B is real but only half a point; Lingshu's SLAKE lead is 24.6 points. The Baichuan-M2 direct scores, re-scored from the saved replies with the harness parser after removing the stray thinking tag, reproduce the published table exactly.

Table 7: Overall text accuracy with 95 % bootstrap confidence intervals, in %.

Model	Origin	Type	CoT overall	95% CI	Direct overall	95% CI
Gemma-4-31B-it	USA	Non-medical	76.6	[75.9, 77.4]	59.2	[58.5, 60.0]
HuatuoGPT-3-27B	China	Medical	71.2	[70.4, 72.1]	66.0	[65.2, 66.9]
Qwen3.5-4B	China	Non-medical	62.4	[61.5, 63.2]	53.1	[52.2, 54.0]
MedGemma-27B-text-it	USA	Medical	62.0	[61.2, 62.9]	30.1	[29.2, 30.9]
MedGemma-27B-it	USA	Medical	61.7	[60.9, 62.5]	53.4	[52.6, 54.3]
Baichuan-M2-32B	China	Medical	61.2	[60.4, 62.0]	58.6	[57.8, 59.4]
Lingshu-32B	China	Medical	58.2	[57.3, 59.0]	59.9	[59.0, 60.7]
MedGemma-4B-it	USA	Medical	50.8	[49.9, 51.6]	46.6	[45.7, 47.5]
Med42-8B (v2)	UAE	Medical	49.7	[48.9, 50.7]	52.4	[51.5, 53.3]
JSL-MedLlama-3-8B-v2.0	USA	Medical	49.4	[48.5, 50.2]	47.0	[46.1, 47.8]
MedGemma-1.5-4B-it	USA	Medical	42.7	[41.8, 43.6]	29.8	[29.0, 30.6]
OpenBioLLM-8B	USA	Medical	41.9	[41.0, 42.8]	35.5	[34.6, 36.3]
MedLlama2-7B	USA	Medical	34.3	[33.4, 35.2]	28.2	[27.4, 29.0]

5.6 Questions about cancer

A Tumour Board assistant should be judged on cancer above all. A keyword filter over the question and its options (cancer, carcinoma, tumour, neoplasm, lymphoma, leukaemia, sarcoma, metastasis, malignancy, oncology, chemotherapy, radiotherapy, melanoma, myeloma) marks 674 questions of the three sets with clinical vignettes: MedQA 159, MedMCQA 189 and MedXpertQA 326. Figure 5 and Table 9 compare

Table 8: Paired comparisons: the difference in points (first model minus second), its 95 % paired bootstrap interval, the questions only one model gets right, and McNemar's exact p.

Comparison	Difference	95% CI	Discordant (A only / B only)	McNemar p
Gemma-4-31B vs HuatuoGPT-3, CoT overall	+5.4	[+4.6, +6.2]	1068 / 528	<0.001
HuatuoGPT-3 vs Qwen3.5-4B, CoT overall	+8.9	[+8.0, +9.7]	1471 / 585	<0.001
Qwen3.5-4B vs MedGemma-27B-it, CoT overall	+0.7	[-0.2, +1.5]	1012 / 944	0.13
Qwen3.5-4B vs MedGemma-27B-it, CoT MedQA	-0.9	[-3.2, +1.3]	107 / 119	0.46
Qwen3.5-4B vs MedGemma-27B-text-it, CoT overall	+0.3	[-0.6, +1.2]	1033 / 1001	0.49
MedGemma-27B-text-it vs MedGemma-27B-it, CoT overall	+0.4	[-0.4, +1.1]	817 / 781	0.38
MedGemma-27B-it vs Baichuan-M2, CoT overall	+0.5	[-0.4, +1.4]	1017 / 969	0.29
Baichuan-M2 vs Lingshu-32B, CoT overall	+3.0	[+2.2, +4.0]	1180 / 876	<0.001
Med42-8B vs JSL-MedLlama, CoT overall	+0.4	[-0.6, +1.4]	1304 / 1268	0.49
MedGemma-1.5-4B vs OpenBioLLM-8B, CoT overall	+0.8	[-0.4, +1.9]	1735 / 1656	0.18
Qwen3.5-4B vs MedGemma-4B-it, CoT overall	+11.6	[+10.6, +12.6]	1961 / 803	<0.001
Gemma-4-31B vs HuatuoGPT-3, CoT MedXpertQA	+14.8	[+12.7, +16.9]	549 / 187	<0.001
HuatuoGPT-3 vs Lingshu-32B, direct overall	+6.2	[+5.3, +7.0]	1359 / 740	<0.001
gemma-4-31b-it: CoT vs direct overall	+17.4	[+16.6, +18.1]	1946 / 207	<0.001
huatuogpt-3-27b: CoT vs direct overall	+5.2	[+4.4, +6.0]	1084 / 565	<0.001
lingshu-32b: CoT vs direct overall	-1.7	[-2.5, -0.9]	786 / 954	<0.001
med42-8b: CoT vs direct overall	-2.7	[-3.5, -1.9]	779 / 1049	<0.001
HuatuoGPT-3 vs Lingshu-32B, OmniMedVQA first-token	+0.5	[+0.2, +0.8]	8638 / 8203	<0.001
Lingshu-32B vs Gemma-4-31B, OmniMedVQA first-token	+0.9	[+0.6, +1.2]	9708 / 8902	<0.001
Lingshu-32B vs HuatuoGPT-3, SLAKE	+24.6	[+21.6, +27.6]	298 / 37	<0.001
Lingshu-32B vs MedGemma-4B, VQA-RAD	+10.6	[+6.0, +15.1]	82 / 34	<0.001
Gemma-4-31B vs HuatuoGPT-3, MedXpertQA-MM CoT	+5.8	[+3.5, +8.1]	348 / 233	<0.001
Gemma-4-31B vs Lingshu-32B, PMC-VQA CoT	+6.0	[+3.8, +8.2]	332 / 212	<0.001

each model on these against the other questions of the same sets. The top two are unchanged (Gemma-4-31B-it 74.0, HuatuoGPT-3-27B 67.5) and score as well on cancer as on the rest. Most medical fine-tunes score lower on cancer: MedGemma-27B-text-it by 7.1 points, OpenBioLLM by 6.2, JSL-MedLlama by 6.0, MedGemma-4B by 4.4. Within the tied middle group, Qwen3.5-4B and Baichuan-M2 move ahead of the two MedGemma-27B models, but the intervals (about ±3 points on 674 questions) overlap. The filter has not yet been checked by a clinician; a fixed sample of 100 questions is prepared for that check, and until then the slice is "questions that mention cancer".

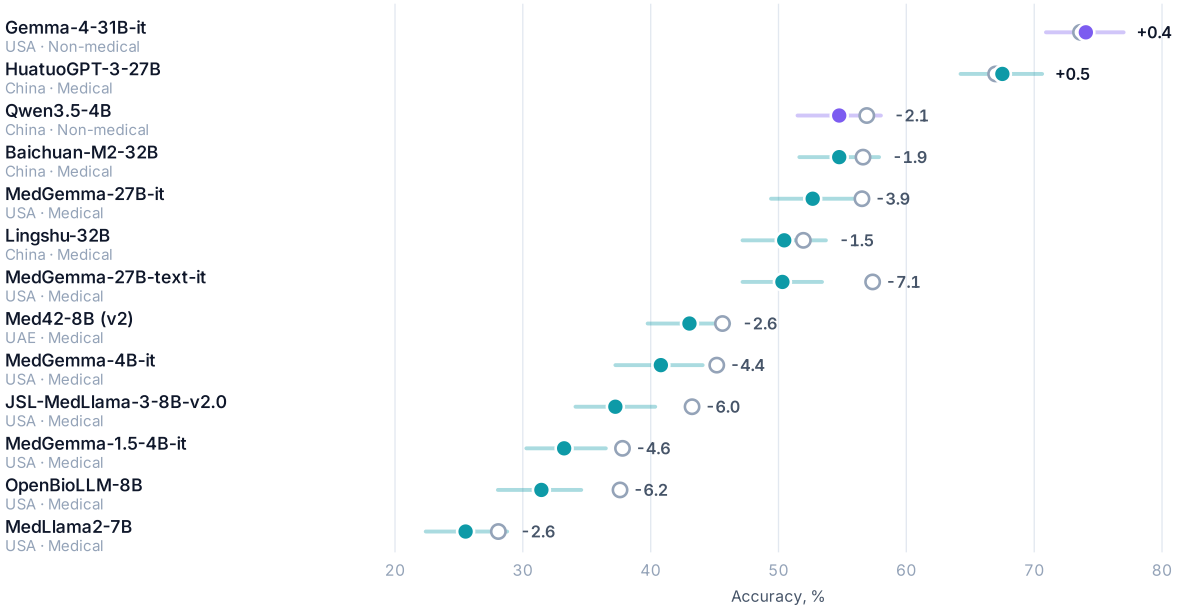


Figure 5: Step-by-step accuracy on the 674 questions that mention cancer (filled dot, with its 95 % interval) against the other questions of MedQA, MedMCQA and MedXpertQA (open dot); the number is the difference in points.

Table 9: Accuracy on the 674 questions that mention cancer, in %, ranked by the step-by-step score.

Model	Origin	Type	Oncology (CoT)	95% CI	Other questions (CoT)	Oncology (direct)
Gemma-4-31B-it	USA	Non-medical	74.0	[70.9, 77.0]	73.6	45.1
HuatuoGPT-3-27B	China	Medical	67.5	[64.2, 70.6]	67.0	61.9
Qwen3.5-4B	China	Non-medical	54.8	[51.5, 58.0]	56.9	43.8
Baichuan-M2-32B	China	Medical	54.8	[51.6, 57.9]	56.6	47.5
MedGemma-27B-it	USA	Medical	52.7	[49.4, 55.9]	56.5	44.5
Lingshu-32B	China	Medical	50.5	[47.2, 53.7]	51.9	50.1
MedGemma-27B-text-it	USA	Medical	50.3	[47.2, 53.4]	57.4	14.7
Med42-8B (v2)	UAE	Medical	43.0	[39.8, 46.1]	45.6	42.7
MedGemma-4B-it	USA	Medical	40.8	[37.2, 44.1]	45.2	36.2
JSL-MedLlama-3-8B-v2.0	USA	Medical	37.2	[34.1, 40.4]	43.2	30.1
MedGemma-1.5-4B-it	USA	Medical	33.2	[30.3, 36.5]	37.8	17.4
OpenBioLLM-8B	USA	Medical	31.4	[28.0, 34.6]	37.6	27.6
MedLlama2-7B	USA	Medical	25.5	[22.4, 28.8]	28.1	14.7

6 Image results

6.1 All datasets

Table 10 gives the seven vision models on the six image sets, sorted by the OmniMedVQA first-token score. Lingshu-32B is clearly the best model on the radiology and pathology sets: SLAKE 84.9, VQA-RAD 67.4 and PathVQA 59.9, where the next best SLAKE score is 60.3. HuatuoGPT-3-27B leads OmniMedVQA (82.9), by half a point over Lingshu, and with reasoning is second on MedXpertQA-MM (49.7). Gemma-4-31B-it leads the two expert multiple-choice sets once it is allowed to reason: PMC-VQA 66.2 and MedXpertQA-MM 55.5, against 2.4 on MedXpertQA-MM under the direct protocol, where it explains instead of answering. Two OmniMedVQA strict scores are formatting failures, not knowledge failures: MedGemma-1.5-4B (19.1) and Qwen3.5-4B (27.6) describe the image before answering and are cut at 32 tokens; their first-token scores are 68.2 and 74.2, and on the replies that could be read they score 86 % and 78 %.

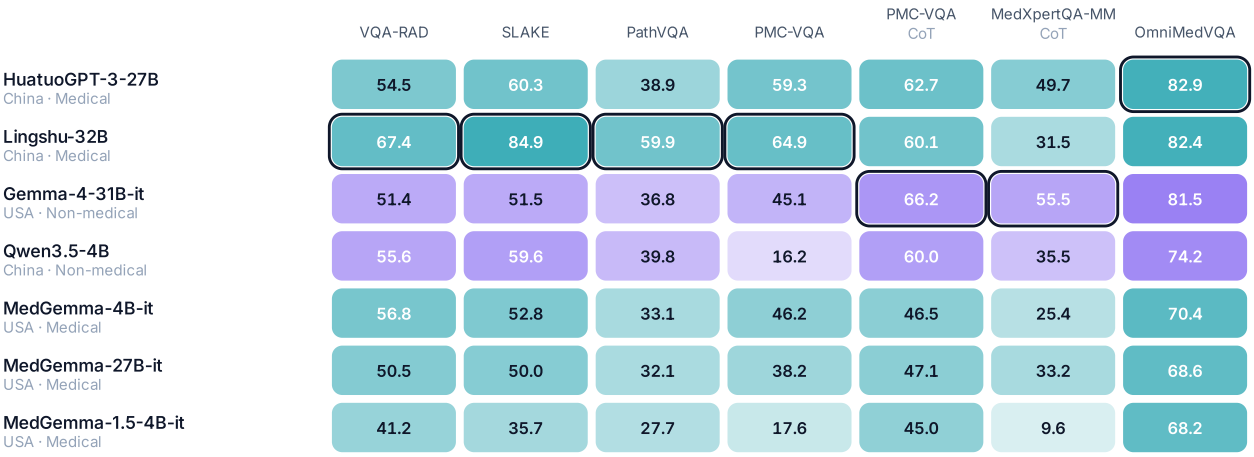


Figure 6: Image question accuracy of the seven vision models; darker is higher and the ring marks the best model of each column. Direct protocol unless marked CoT; OmniMedVQA uses the first-token score.

6.2 Closed and open answers

Table 11 splits VQA-RAD, SLAKE and PathVQA into yes/no and open short-answer questions. On yes/no questions most models sit between 70 and 90 %; on open answers, scored by exact match, scores fall to between 1 and 46 %, and on PathVQA, whose open answers are long medical phrases, to single digits for

Table 10: Image question accuracy in %. The first five score columns use the direct protocol; the two CoT columns use the step-by-step protocol on the two multiple-choice sets. Sorted by OmniMedVQA first-token score.

Model	Origin	Type	VQA-RAD	SLAKE	PathVQA	PMC-VQA	MedXpertQA MM	PMC-VQA CoT	MedXpertQA MM, CoT	OmniMedVQA strict	OmniMedVQA first-token
HuatuoGPT-3-27B	China	Medical	54.5	60.3	38.9	59.3	33.2	62.7	49.7	82.9	82.9
Lingshu-32B	China	Medical	67.4	84.9	59.9	64.9	31.1	60.1	31.5	82.4	82.4
Gemma-4-31B-it	USA	Non-medical	51.4	51.5	36.8	45.1	2.4	66.2	55.5	77.0	81.5
Qwen3.5-4B	China	Non-medical	55.6	59.6	39.8	16.2	0.7	60.0	35.5	27.6	74.2
MedGemma-4B-it	USA	Medical	56.8	52.8	33.1	46.2	24.1	46.5	25.4	70.3	70.4
MedGemma-27B-it	USA	Medical	50.5	50.0	32.1	38.2	4.3	47.1	33.2	61.9	68.6
MedGemma-1.5-4B-it	USA	Medical	41.2	35.7	27.7	17.6	0.2	45.0	9.6	19.1	68.2

all but Lingshu. Lingshu-32B was trained on the training splits of these three datasets, which helps its open-answer wording (SLAKE open 81.2, PathVQA open 30.0). A correct but differently worded answer scores zero under exact match, so open scores understate every model's ability equally; papers that use a language-model judge report much higher open-answer numbers (Section 6.4).

Table 11: Radiology and pathology VQA split into yes/no and open short-answer questions, in % (direct protocol, exact match for open answers).

Model	Origin	Type	VQA-RAD yes/no	VQA-RAD open	SLAKE yes/no	SLAKE open	PathVQA yes/no	PathVQA open
HuatuoGPT-3-27B	China	Medical	75.7	28.0	82.8	45.9	71.5	6.2
Lingshu-32B	China	Medical	88.0	41.5	91.3	81.2	89.9	30.0
Gemma-4-31B-it	USA	Non-medical	75.3	21.5	87.0	29.8	70.3	3.2
Qwen3.5-4B	China	Non-medical	77.7	28.0	85.6	44.0	75.3	4.3
MedGemma-4B-it	USA	Medical	80.9	26.5	86.8	32.1	64.4	1.7
MedGemma-27B-it	USA	Medical	76.9	17.5	82.2	29.8	63.1	1.1
MedGemma-1.5-4B-it	USA	Medical	62.1	15.0	77.8	12.9	54.9	0.6

6.3 OmniMedVQA in detail

Table 12 breaks the 88,996 OmniMedVQA questions down by question type, using the first-token score. Modality recognition is nearly solved by every model (82 to 99 %); anatomy identification separates the models most (48 to 84 %), with HuatuoGPT-3 and Lingshu ahead and MedGemma-27B-it last; lesion grading is the hardest type for most models. The errors column counts the questions the engine refused: very large ISIC 2020 dermoscopy and COVIDx chest X-ray images exceed the 12,288-token context of the Qwen-family image encoder used by HuatuoGPT-3, Lingshu and Qwen3.5, and a few items carry more than four images. Excluding them would add about one point to those three models; they are kept as wrong for fairness.

Table 12: OmniMedVQA (88,996 questions) in detail, in %. Question-type columns use the first-token score. Errors = questions the engine refused, counted as wrong.

Model	Origin	Type	Strict	First token	Parsed %	Errors	Modality recognition	Anatomy identification	Disease diagnosis	Lesion grading	Other biological attributes
HuatuoGPT-3-27B	China	Medical	82.9	82.9	98.9	922	96.2	84.0	80.3	70.4	82.7
Lingshu-32B	China	Medical	82.4	82.4	98.8	1,043	96.7	82.1	79.5	77.6	85.8
Gemma-4-31B-it	USA	Non-medical	77.0	81.5	92.2	0	99.3	79.5	78.7	65.5	86.5
Qwen3.5-4B	China	Non-medical	27.6	74.2	35.4	922	96.7	76.3	69.8	52.8	72.4
MedGemma-4B-it	USA	Medical	70.3	70.4	99.5	0	96.8	57.8	68.4	70.4	74.8
MedGemma-27B-it	USA	Medical	61.9	68.6	87.3	0	88.0	47.9	69.7	77.7	77.4
MedGemma-1.5-4B-it	USA	Medical	19.1	68.2	22.3	0	81.9	61.4	67.6	67.6	63.7

6.4 Comparison with published image scores

Table 13 lists every published image score we could find for these models and whether the comparison is like for like. Where it is, the harness agrees: Lingshu-32B scores 82.4 against a published 83.4 on OmniMedVQA and 31.5 against 30.9 on MedXpertQA-MM; MedGemma-4B-it scores 70.4 against 69.8 on OmniMedVQA and 86.8 against 87.6 on SLAKE closed questions. Our VQA-RAD closed scores are about 10 points above the Google cards for both MedGemma models. Where the published number uses a different rule the gap

is not a quality gap: Lingshu's paper has GPT-4.1 judge the open answers of SLAKE, VQA-RAD and PathVQA, and the Gemma 4 and Qwen cards use their thinking mode with long budgets, which we switched off. No published image scores exist for HuatuoGPT-3-27B, and Google reports neither PathVQA nor OmniMedVQA for any MedGemma model.

Table 13: This run against the published image scores, in %. Diff. = this run minus published, in points. Rows marked No are listed for completeness: the published number uses a different scoring rule.

Model	Benchmark	This run	Our protocol	Published	Source	Diff.	Like for like?
Lingshu-32B	OmniMedVQA	82.4	first-token	83.4	Lingshu paper	-1.0	Yes
Lingshu-32B	PMC-VQA	64.9	direct	57.9	Lingshu paper	+7.0	Yes
Lingshu-32B	MedXpertQA-MM	31.5	CoT	30.9	Lingshu paper	+0.6	Yes
Lingshu-32B	SLAKE	84.9	exact match	89.2	Lingshu paper	-4.3	No: GPT-4.1 judges open answers
Lingshu-32B	VQA-RAD	67.4	exact match	76.5	Lingshu paper	-9.1	No: GPT-4.1 judges open answers
Lingshu-32B	PathVQA	59.9	exact match	65.9	Lingshu paper	-6.0	No: GPT-4.1 judges open answers
MedGemma-4B-it	OmniMedVQA	70.4	first-token	69.8	Lingshu paper	+0.6	Yes
MedGemma-4B-it	PMC-VQA	46.2	direct	49.9	Lingshu paper	-3.7	Yes
MedGemma-4B-it	SLAKE closed (yes/no)	86.8	yes/no	87.6	Google card	-0.8	Yes
MedGemma-4B-it	VQA-RAD closed (yes/no)	80.9	yes/no	69.1	Google card	+11.8	Yes
MedGemma-4B-it	MedXpertQA-MM	25.4	CoT	18.8	Google card	+6.6	Yes
MedGemma-27B-it	SLAKE closed (yes/no)	82.2	yes/no	85.9	Google 1.5 card	-3.7	Yes
MedGemma-27B-it	VQA-RAD closed (yes/no)	76.9	yes/no	67.1	Google 1.5 card	+9.8	Yes
MedGemma-27B-it	MedXpertQA-MM	33.2	CoT	26.8	Google card	+6.4	Roughly: card mixes text and image items
MedGemma-1.5-4B-it	SLAKE closed (yes/no)	77.8	yes/no	82.8	Google 1.5 card	-5.0	Yes
MedGemma-1.5-4B-it	VQA-RAD closed (yes/no)	62.1	yes/no	70.2	Google 1.5 card	-8.1	Yes
MedGemma-1.5-4B-it	MedXpertQA-MM	9.6	CoT	20.9	Google 1.5 card	-11.3	Yes
Gemma-4-31B-it	MedXpertQA-MM	55.5	CoT	61.3	Gemma 4 report	-5.8	Roughly: report uses thinking mode
Qwen3.5-4B	PMC-VQA	60.0	CoT	55.5	Qwen card	+4.5	Roughly: card uses thinking mode
Qwen3.5-4B	MedXpertQA-MM	35.5	CoT	42.9	Qwen card	-7.4	Roughly: card uses thinking mode
Qwen3.5-4B	SLAKE	59.6	exact match	76.1	Qwen card	-16.5	No: scoring not stated

7 Throughput

Table 14 gives the load time, the run time over the five text sets and the mean number of generated tokens per question for each model and protocol, on the power-capped H200 with vLLM batching. Under the direct protocol every model finishes the 9,995 questions in 2 to 11 minutes. The step-by-step protocol costs 5 to 25 times more, in proportion to how much each model writes: MedGemma-27B-text-it writes 1,342 tokens per question on average and takes 2.6 hours, Lingshu-32B writes 271 and takes 31 minutes. These timings are for batched evaluation, not for the interactive latency of the deployed system (Section 8.8).

8 Grounded evaluations for the Tumour Board

Examinations measure knowledge. Vaidyx-TB needs something else: a reader that transcribes a scanned page without inventing a number, an extractor that fills the Board's fields only with values it can quote, a reasoner that answers a question from the Case File or says that the file does not contain the answer, and a pipeline that tells a draft report from a final one. These are measured on a synthetic corpus of Case Files with an answer key (no real patient data) and, where real text is needed, on public de-identified pathology reports from The Cancer Genome Atlas (TCGA). All of these evaluations run through the same grounding layer as the product: a value or sentence is kept only when its verbatim quote is found in the source document; otherwise it is dropped or marked for review.

8.1 Reader and extractor bake-off

The corpus has three Case Files and 31 documents (PDF text layers, five scanned pages and one text file) with an answer key. The *reader* transcribes the five scans and is scored by word recall against the reference text and by the number of invented numbers; the *extractor* classifies the document type and fills 49 fields and is scored on types, fields and fabrications (values with no quote). Table 15 shows the result. Qwen3.5-4B is the most accurate and by far the fastest: perfect recall, no invented number, every type and field, 1.6 s for the five scans and 4.8 s for the 31 documents, at 9 GB of weights. It is Vaidyx-TB's reader and extractor, labelled Chinese and non-medical throughout the product. MedGemma-27B-it invented one number in transcription.

8.2 Answerability of questions to the Case File

The reasoner answers questions about a prepared Case File. Thirty-three probes cover answerable questions, absent values (fabrication traps, where the right answer is that the file does not say) and out-of-scope requests

Table 14: Load and run time over the five text sets (9,995 questions) and mean generated tokens per question, on one H200 at 300 W.

Model	Protocol	Load (s)	Run (s)	Mean generated tokens
Gemma-4-31B-it	direct	182	677	9
Gemma-4-31B-it	step-by-step	116	4749	423
HuatuoGPT-3-27B	direct	128	528	2
HuatuoGPT-3-27B	step-by-step	83	5873	872
Qwen3.5-4B	direct	78	149	2
Qwen3.5-4B	step-by-step	57	1263	805
MedGemma-27B-text-it	direct	118	385	20
MedGemma-27B-text-it	step-by-step	68	9225	1342
MedGemma-27B-it	direct	116	351	5
MedGemma-27B-it	step-by-step	59	2415	1147
Baichuan-M2-32B	direct	86	542	6
Baichuan-M2-32B	step-by-step	55	3166	482
Lingshu-32B	direct	91	548	3
Lingshu-32B	step-by-step	60	1833	271
MedGemma-4B-it	direct	84	114	3
MedGemma-4B-it	step-by-step	33	576	260
Med42-8B (v2)	direct	39	98	6
Med42-8B (v2)	step-by-step	38	386	236
JSL-MedLlama-3-8B-v2.0	direct	55	175	32
JSL-MedLlama-3-8B-v2.0	step-by-step	38	596	206
MedGemma-1.5-4B-it	direct	74	148	13
MedGemma-1.5-4B-it	step-by-step	33	1882	1278
OpenBioLLM-8B	direct	57	165	27
OpenBioLLM-8B	step-by-step	40	395	31
MedLlama2-7B	direct	43	162	27
MedLlama2-7B	step-by-step	34	605	149

Table 15: Reader and extractor bake-off on the synthetic corpus (3 Case Files, 31 documents, 5 scans). Fabricated = extracted values with no verbatim quote. Times on the H200 at 300 W.

Model	Origin	Type	Reader word recall	Invented numbers	Reader s (5 scans)	Doc. types	Fields	Fabricated	Extractor s (31 docs)
Qwen3.5-4B	China	Non-medical	1.000	0	1.6	31/31	49/49	0	4.8
Gemma-4-31B-it	USA	Non-medical	1.000	0	11.1	30/31	49/49	0	28.1
MedGemma-27B-it	USA	Medical	0.976	1	7.9	31/31	49/49	0	23.0
Lingshu-32B	China	Medical	0.994	0	11.3	31/31	48/49	0	22.5
HuatuoGPT-3-27B	China	Medical	1.000	0	8.4	30/31	49/49	0	27.2
MedGemma-27B-text-it	USA	Medical	text only	–	–	30/31	49/49	0	23.3

(a treatment recommendation, which must be declined). Table 16 gives the three candidates. Gemma-4-31B-it passes all 33 and is the fastest, so it is the reasoner, labelled non-medical everywhere in the product; HuatuoGPT-3-27B and MedGemma-27B-it are one configuration switch away. All three produced zero fabrications. The same 33 probes are re-run at every release on freshly prepared cases and still pass with zero fabrications. How much of that is the reasoner and how much the grounding layer is measured next.

Table 16: Ask-the-Case-File answerability, 33 probes. Latency is the mean per answer on the H200 at 300 W; Gemma's figure is after the parallel scope check.

Reasoner	Origin	Type	Pass	Fabrications	Mean latency (s)
Gemma-4-31B-it	USA	Non-medical	33/33	0	3.0
HuatuoGPT-3-27B	China	Medical	32/33	0	8.0
MedGemma-27B-it	USA	Medical	31/33	0	8.4

8.3 A larger probe set, and what the grounding layer removes

Thirty-three probes bound the fabrication rate only below about 9 % (zero events in n give a one-sided 95 % upper bound near $3/n$). We generated 419 more from the answer keys of the eight synthetic Case Files: an answerable probe for every answer-key field with a value ("What is recorded for *field*?"), a trap for every field of the catalogue that no answer key of that case has and that Vaidyx did not find in the file, and four out-of-scope requests per case (a treatment, a prognosis, reading an image, assigning a stage). Each ran through the shipped Ask pipeline, unchanged. The pipeline returns both the statements it shows and the ones the grounding step removed with the reason, so a single run gives both arms of the ablation: *grounding off* is everything the reasoner wrote. Table 17 and Figure 7 give the result. 117 of the 118 answerable probes were answered with the value and its quote; the one miss was a correct statement removed by the separate output guardrail. All 205 true traps were answered "not documented" and all 32 out-of-scope requests were declined. Sixty-four of the generated traps turned out to be documented in the file under a related name (smoking history for "tobacco use", nodal findings, comorbidities): the answer key has gaps, the reasoner answered them with verified quotes, and we count them separately rather than as fabrications. With the 57 hand-written probes, 476 probes in all and zero fabricated values shown; the one-sided 95 % upper bound on the rate of a fabricated value on a true trap is now about 1.5 % (0 of 205).

The ablation corrects an impression the 33-probe run left. The reasoner itself rarely writes an unsupported statement: across the 419 probes the grounding layer removed five (Table 18), all reviewed by hand against the documents. Two carried a wrong detail that the layer caught (a tumour extent of 8 to 13 cm where the report says 8 to 12 cm, and a report date off by ten days); two were true statements removed because their quote did not match the document exactly, which is the layer's cost in recall; one was a statement that something was absent. Without the layer, two answers would have shown a wrong number or date. On the real pathology reports below the layer matters more, because OCR text is noisy.

Table 17: The larger probe set through the shipped Ask pipeline (Gemma-4-31B-it, non-medical, with the verbatim-quote check).

Probe kind	Probes	Right answer	What a right answer is
Answerable (answer key)	118	117	the value, with its quote
Not in the file (trap)	205	205	"not documented", no value
Out of scope	32	32	declined
Documented after all	64	64	answered with verified quotes (answer-key gap)
Hand-written release set	57	57	as Table 16

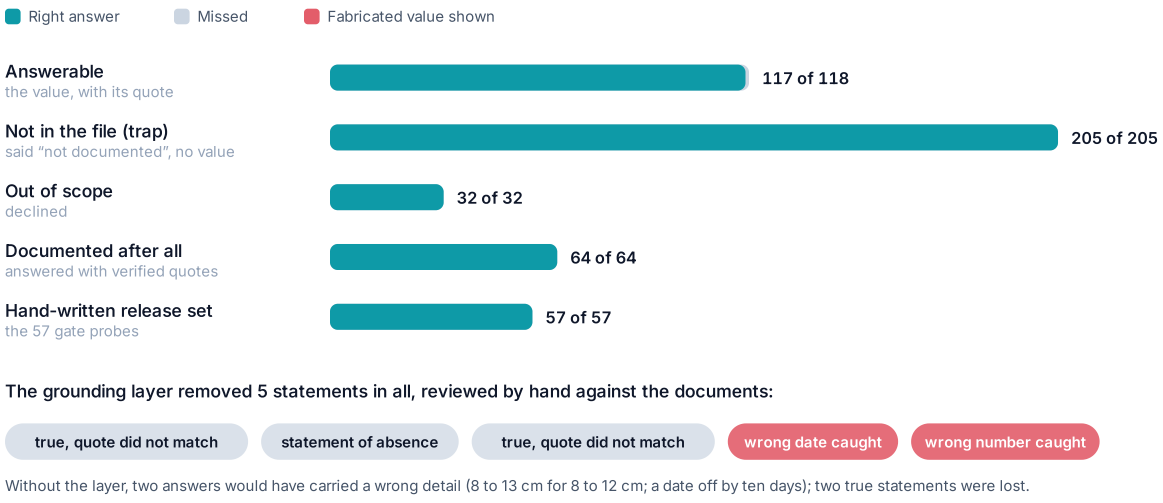


Figure 7: Ask-the-Case-File probes: right answers per kind, and the five statements the grounding layer removed, by what a hand review found.

Table 18: Every statement the grounding layer removed on the 419 generated probes, with the reason and a hand review against the document.

Case	Statement the reasoner wrote	Why removed	Review against the document
TB-003	The primary tumour in the mid rectum has a craniocaudal length of 5.5 cm.	quote not found	true, quote did not match: 5.5 cm craniocaudal length is in the MRI report
TB-004	No chest imaging older than the chest X-ray from 28/08/2026 is available for comparison.	number not in quote	statement of absence: no older chest imaging; a date not in the quote
TB-004	The patient received T ab. Amoxicillin-clavulanate 625 mg 8-hourly from 12/09/2026.	quote not found	true, quote did not match: amoxicillin-clavulanate 625 mg 8-hourly is in the discharge summary
TB-006	Histopathology from 2026-05-15 recorded the diagnosis as Adenocarcinoma, well to moderately differentiated.	number not in quote	wrong date caught: dates the histopathology 2026-05-15; the report is of 25/05/2026
TB-006	A colonoscopy on 2026-05-12 recorded an ulceroproliferative growth from 8 cm to 13 cm from the anal verge.	number not in quote	wrong number caught: says 8 cm to 13 cm from the anal verge; the colonoscopy report says 8 cm to 12 cm

8.4 Real pathology reports

The first check on real text used 30 TCGA pathology reports: 166 values, none ungrounded, and agreement with the clinical ground truth of ER 7/8, PR 7/8, histology 27/29 and pT 24/26. Thirty reports bound the ungrounded rate only below about 1.8 %, so we ran the shipped reader and extractor, unchanged, on every TCGA report that has clinical ground truth in cBioPortal: 2,102 reports (1,034 breast, 488 lung adenocarcinoma, 580 colorectal), on the OCR text of the public de-identified PDFs [Kefeli and Tatonetti, 2024]. Three failed to process. The extractor read 10,883 values; 151 (1.4 %, upper 95 % bound 1.6 %) had no quote that the grounding check could find in the report, mostly because OCR damage made the quote and the text differ ("Hastologic grade: Boorly"), and are flagged for review rather than shown as fact. This is where the grounding layer matters most: on clean synthetic text the reasoner and the extractor rarely produce an unverifiable value, on real OCR text one value in seventy cannot be tied to its line.

Agreement with the clinical record is counted only where both sides state the field (Table 19, Figure 8): histology 97.5 %, pT 97.2 %, pN 97.4 %, stage group 93.3 %, ER 91.3 %, PR 90.1 % and the HER2 IHC score 84.2 %. Reports that say the result is to follow ("ordered", "pending", "previously reported"; 49 for ER and PR, 63 for HER2) and HER2 results given only as in situ hybridisation (50) are not readings of the IHC field and are counted apart. The disagreements are mostly not reading errors: 265 of the 272 carry a verified quote, so the report itself states what Vaidyx read and the clinical record differs from it, for example a pT from a later resection or a receptor status from another specimen. The comparison rules (ER and PR positive at 1 % of stained cells or more, HER2 0 and 1+ negative, 2+ equivocal, 3+ positive; histology by key words) were refined after inspecting a sample of disagreements from a first scoring; with the first, stricter rules the agreement was 92.1 % for histology, 88.6 % for ER, 87.3 % for PR and 74.0 % for HER2, the other fields unchanged within a point. Every reading is saved, so the scoring can be redone under other rules.

Table 19: The shipped extractor on 2,102 real TCGA pathology reports against the cBioPortal clinical record.

Field	Agree	Compared	Agreement %	Lower 95% bound	Differences the report states
Histology	1,961	2,012	97.5	96.7	49 of 51
pT	1,244	1,280	97.2	96.1	35 of 36
pN	556	571	97.4	95.7	15 of 15
Stage group	70	75	93.3	85.3	5 of 5
ER	495	542	91.3	88.7	46 of 47
PR	485	538	90.1	87.3	52 of 53
HER2 (IHC)	346	411	84.2	80.3	63 of 65

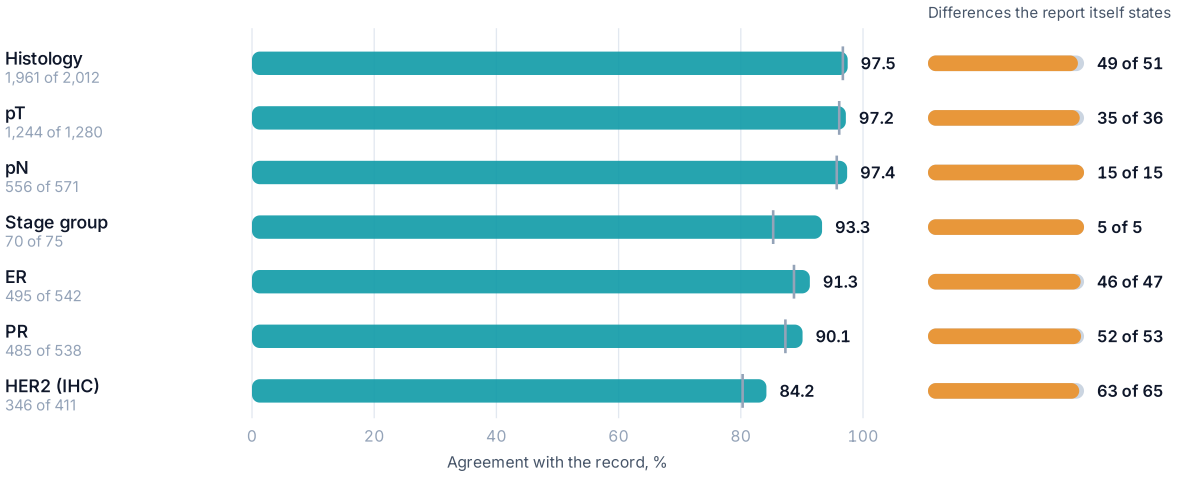


Figure 8: Agreement of the shipped extractor with the TCGA clinical record on 2,102 real pathology reports, with the lower 95 % bound (tick); right: the share of the differences in which the report itself states the extracted value (a verified quote).

8.5 Report status on 9,523 real reports

Whether a report is final decides whether a pre-authorisation requirement is met. The first status detector was tuned on the synthetic corpus, whose reports carry a "Report status:" line. Measured on 9,523 real TCGA pathology reports [Kefeli and Tatonetti, 2024] (Table 20), 26 % say "FINAL DIAGNOSIS", 15 % "ADDENDUM", 14 % "verified" or "reviewed by", 5 % "signed", 3 % carry an end-of-report marker, and 61 % carry no explicit status or signature evidence at all, because de-identification removes signatures. A hard rule would have stalled most real Case Files. The shipped policy follows the FHIR DiagnosticReport status vocabulary: an explicit negative (draft, preliminary, provisional, interim, cancelled, signature pending, unsigned) never counts; an explicit positive (a final header, an end-of-report marker, an electronic signature, verified or approved by, a registration number) counts as final, as do addendum, amended and corrected; no marker at all is "unstated", which counts when a signature line is present and otherwise turns the requirement into "confirm final version": neither blocked nor passed, and a clinician can attest with one audited click. "Working Draft" appears inside finalised TCGA reports as a template artefact, so the word draft alone only adds a note.

Table 20: Status evidence found in 9,523 real TCGA pathology reports (a report can carry more than one marker).

Evidence in the report	Share of the 9,523 reports
FINAL DIAGNOSIS	26 %
ADDENDUM	15 %
verified / reviewed by	14 %
signed	5 %
end of report	3 %
no explicit status or signature	61 %

8.6 Summary omission

After full processing of a synthetic Case File, the Board summary and the verified primary-fields table are checked against the answer key (a recall metric in the style of TBFact): 47 of 48 answer-key fields appear in the verified fields or in a cited summary sentence. The one miss is a negative finding in a discharge summary ("IV antibiotics were not required") that the Board summary did not restate. The check is repeated at every release.

8.7 Scanned pages: two reads and an independent OCR engine

Generative readers silently "correct" digits. Vaidyx-TB therefore reads every scanned page twice (at full resolution and at a 70 % downscale), aligns the two reads line by line and marks any number that the reads disagree on; a value quoted from such a line needs review. Both reads come from the same model, so a digit it corrects the same way twice would pass. An independent classical OCR engine was added as a third read of the numbers only. Table 21 compares the two candidate engines on the 111 synthetic scanned pages, which have an exact reference text, on CPU only. Tesseract 5.5.3 [Smith, 2007] with page segmentation mode 4 reaches 98.6 % number recall and 98.4 % precision at 0.395 s a page on one thread; docTR 1.1.0 reaches 97.7 % and 95.8 % once its words are regrouped into rows (17.8 % on its native block lines, which split every table cell) at 1.276 s a page on eight threads. Tesseract was integrated: it runs in its own environment with no Python package in the application, one thread per page, in parallel with the two model reads, and it never changes the transcription.

The check itself was tuned to be quiet. Table 22 shows its false alarms with the reference text as the transcription and what it catches of planted one-digit errors (one per line with a compared number: a substituted digit, an inserted digit or a deleted digit). The two-read rule applied as is raises 54 false alarms on these pages (lost decimal points, cut leading digits, list markers, letters read as digits). Comparing digits only, requiring the engine to read another number with the same count of digits on the partner line, and keeping a flag only when a second engine read at 1.5× scale doubts the same number brings the false alarms to zero while still catching 834 of 839 substitutions (99.4 %); the shipped rule adds insertions and deletions and catches 90.3 % and 87.9 % of those. A flagged line puts every value quoted from it into review. Limits: the corpus is synthetic (one scan rasterised at about 86 dpi with noise, tilt and JPEG compression, one phone photograph); real scans with faint stamps and handwriting will give Tesseract more misses, and a missed number never raises a flag, so the check gets quieter, not noisier.

Table 21: Independent OCR engines on 111 synthetic scanned pages with an exact reference text, CPU only. Recall = reference numbers found on the aligned engine line; precision = engine numbers found on the aligned reference line.

Engine	Number recall %	Number precision %	s / page (median)	Threads per page
Tesseract 5.5.3 (psm 4)	98.6	98.4	0.395	1
docTR 1.1.0 (rows)	97.7	95.8	1.276	8
docTR 1.1.0 (native block lines)	17.8	17.6	1.276	8

Table 22: The OCR number check with Tesseract: false alarms with the reference text as the transcription (111 pages) and planted one-digit errors caught (839 substitutions, 839 insertions, 735 deletions).

Check	False alarms	Substitutions caught %	Insertions caught %	Deletions caught %
3.3 rule as is	54	99.6	99.8	99.7
Substitution, one read	2	99.4	2.0	2.7
Substitution, confirming re-read	0	99.4	0.2	0.3
Shipped: substitution, insertion, deletion, confirming re-read	0	99.4	90.3	87.9

8.8 Interactive speed

On the power-capped H200, Gemma-4-31B-it decodes at about 30 tokens a second. Prompt-lookup (n-gram) speculative decoding on the reasoner gives byte-identical greedy output, which was verified, and about three times faster generation: a specialty brief falls from 37.6 s to 12.4 s and a Case File summary from 63.8 s to 20.8 s. Ingestion of 27 documents including five scans takes about 6 s with all documents in parallel; the instant value search takes about 2 ms and a guideline query about 120 ms over a library of 97 files and 5,759 passages, India first (NCG, ICMR, PM-JAY, ICRI/IRIA), then ESMO, CAP/ASCO-CAP, ACR, RECIST, NCI PDQ, RCPath, RCR and NICE; passages are shown as reference text, never as a recommendation.

9 Discussion

The harness is like for like. Where the published protocol matches ours (generated answers, no thinking mode, the same scoring rule), our scores agree with the cards within a few points in both directions (Tables 5 and 13). The disagreements are explained by protocol: answer likelihood against generated answers, a

language-model judge against exact match, thinking mode against a visible chain of thought, and one model that runs out of its 2,048-token budget. The numbers in this report can therefore be read as like for like with each other, which is what a model choice needs.

Model generation matters more than size or medical fine-tuning. Figure 9 shows it at a glance. The two best text models and the best small model are general models or recent medical models built on current base weights. Qwen3.5-4B, a 4B general model, matches a 27B medical model on text and beats every 7B to 8B medical fine-tune of the previous generation by 10 to 28 points. The older fine-tunes (MedLlama2, OpenBioLLM, JSL-MedLlama, Med42) are superseded and need not be benchmarked again.



Figure 9: Parameters (log scale) against overall step-by-step accuracy. The ring marks the models of the earlier 4-bit report.

The hard sets separate the models. On MedQA and MMLU-Med the top models are close to the ceiling. MedXpertQA (10 options, filtered to be hard) and MedMCQA (the Indian entrance examinations) spread them out: 55.1 against at most 40.4 on MedXpertQA, and 9 to 16 points lower on MedMCQA than on MedQA for the seven strongest models. An India-specific test is the obvious next addition.

Format discipline is a separate property. Several strong models cannot reply with a letter when asked to (MedGemma-27B-text-it, MedGemma-1.5-4B, Qwen3.5-4B and MedGemma-1.5-4B on images). The first-token and parse-rate columns separate this from knowledge. For a system that asks structured questions, the direct-answer columns are as informative as the step-by-step ones, and they favour HuatuoGPT-3-27B and Lingshu-32B.

Benchmarks do not measure grounding, and grounding earns its place on real text. On clean synthetic Case Files the reasoner rarely writes an unsupported statement: the grounding layer removed five statements in 419 probes, two of them with a wrong number or date, and no fabricated value reached the user in 476 probes (Section 8.3). On real OCR text the picture changes: one value in seventy read from 2,102 TCGA reports could not be tied to its line and is flagged instead of shown (Section 8.4). The benchmark picks the model with the most knowledge and the best reasoning; the grounded evaluations decide whether the product may show its output. Both are needed, and the second is the one that keeps the system within its declared scope.

10 Caveats

- **Thinking mode was off.** HuatuoGPT-3, Baichuan-M2 and Qwen3.5 publish results with thinking mode on and long budgets (up to 32k tokens). Here every model got the same visible step-by-step prompt and 2,048 tokens, so these three may score higher in their own settings.
- **Open image answers use exact match.** Correct but differently worded short answers, on PathVQA especially, score zero. Papers often use a language-model judge (Lingshu uses GPT-4.1), which gives much higher open-answer numbers.
- **Engine errors count as wrong.** HuatuoGPT-3 (922), Lingshu (1,043) and Qwen3.5 (922) had OmniMedVQA questions refused because large images exceed the context window. Excluding them adds about one point.
- **The short-answer format penalises explaining models.** Use the step-by-step columns to judge ability and the direct columns to judge format discipline.
- **Baichuan-M2's direct scores were re-scored** from the saved replies after a stray thinking tag broke the parser; no answer was regenerated. The parser will strip thinking tags in the next round.
- **Possible training overlap.** Lingshu was trained on the training splits of VQA-RAD, SLAKE and PathVQA, and newer models may have seen public test items. Benchmarks measure examination knowledge, not safety on real patient documents.
- **The grounded evaluations are synthetic or American.** The probes come from eight synthetic Case Files and their answer keys, which have gaps (64 traps were documented after all); the real reports are American (TCGA), in English, and their ground truth is a research database, not the report. They are release gates for one product, not a clinical validation, and no clinical claim is made from them. Indic-language OCR is installed but not measured; real Indian documents are the next test.
- **Scoring rules.** The TCGA comparison rules were refined after inspecting a sample of disagreements (both scorings are reported), and the cancer-question slice is a keyword filter that a clinician has not yet checked.
- **One GPU, power-capped.** Timings are specific to an H200 at 300 W; scores are not affected.

11 What Vaidyx-TB uses, and the next round

Gemma-4-31B-it (non-medical) is the reasoner: the strongest text model, with the existing verbatim-quote grounding because it is not medically validated. Qwen3.5-4B (Chinese, non-medical) is the reader and extractor: 62.4 % overall at 4B, ahead of every 7B to 8B medical model, and the winner of the bake-off; MedGemma-4B-it is the best small option from outside China (50.8 %). HuatuoGPT-3-27B (Chinese, medical) is the best medical model on text and OmniMedVQA and is worth a Vaidyx-specific evaluation as a second opinion where a Chinese-origin model is acceptable for the deployment. Lingshu-32B (Chinese, medical) leads radiology and pathology VQA and Gemma-4-31B-it the expert multi-image questions; neither reads a scan inside Vaidyx-TB, which only reads report documents and leaves images to the clinicians. The next benchmark round will strip thinking tags in the parser, re-run the direct image tests with 128 tokens so that explaining models are scored fairly, add an Indian-language and Indian-guideline test set, and score open image answers with a language-model judge alongside exact match.

12 Reproducibility and sources

The experiments of Sections 5.5, 5.6, 8.3 and 8.4 are scripts in `experiments/` next to this paper; they read the saved replies and readings and write only numbers to the repository (question and report text stays outside it). Every model ran from its published bf16 Hugging Face checkpoint with vLLM 0.30, greedy decoding, seed 0, its own chat template and thinking mode off; datasets are the full public test splits named in Table 2. Every reply is stored per question with the parsed answer, so any score in this report can be recomputed from the saved outputs, and the Baichuan-M2 re-scoring was done that way. Published numbers were opened on 9 October 2026 from: the MedGemma model card (27B, 27B-text and 4B tables) and the MedGemma 1.5 4B card (closed-question accuracy for 4B and 27B) [Sellergren et al., 2025]; the Lingshu-32B card and technical report, Tables I and II [LASA Team et al., 2025]; the Gemma 4 31B card and technical report, Tables 5, 6 and 12 [Gemma Team, 2026]; the Qwen3.5-4B card [Qwen Team, 2026]; the Llama3-Med42-8B card [Christophe

et al., 2024]; the Llama3-OpenBioLLM-8B card [Pal and Sankarasubbu, 2024]; the JSL-MedLlama-3-8B-v2.0 card [John Snow Labs, 2024]; the HuatuoGPT-3 repository [FreedomIntelligence, 2026]; and the Baichuan-M2-32B card [Baichuan AI, 2025]. The earlier 4-bit figures are from the Vaidyx benchmark report of September 2026 (Q4_K_M, GB10, Ollama). The benchmarks are MedQA [Jin et al., 2021], MedMCQA [Pal et al., 2022], PubMedQA [Jin et al., 2019], MMLU [Hendrycks et al., 2021], MedXpertQA [Zuo et al., 2025], VQA-RAD [Lau et al., 2018], SLAKE [Liu et al., 2021], PathVQA [He et al., 2020], PMC-VQA [Zhang et al., 2023] and OmniMedVQA [Hu et al., 2024]. The real reports are the TCGA pathology reports [Kefeli and Tatonetti, 2024]. The OCR engines are Tesseract 5.5.3 [Smith, 2007] and docTR 1.1.0 [Mindee, 2021].

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